

Model Analysis Report

Building Fire-Resilience Credit Model

MKM-FIRE-001

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Governance Framework: SR 11-7 / SS1/23 Model Risk Management

Contents

1	Sensitivity Analysis	2
2	Stress Testing	5
2.1	Stress Scenarios	5
3	Model Performance	5

1 Sensitivity Analysis

This section presents the sensitivity of model outputs to key input parameters. Parameter perturbation ranges are chosen to cover plausible operating conditions and stress scenarios as required under SR 11-7 guidance.

Table 1: Fire-resilience credit by suppression-systems level (3-storey office baseline)

	Containment (%)	Loss freq (%)	Fire spread (bps)
Suppression: Not assessed	47.0	1.15	8.50
Suppression: Partial	63.1	0.7500	0.0000
Suppression: Meets minimum	66.7	0.6750	0.0000
Suppression: Enhanced	78.1	0.4450	0.0000
Suppression: Verified	86.8	0.2600	0.0000

Table 2: Fire-resilience credit by building height — the fire-service reach cliff (reach = 4 storeys; sub-threshold 'Partial' internal suppression)

	Containment (%)	Loss freq (%)	Fire spread (bps)
1 storeys	63.1	0.7500	0.0000
3 storeys	63.1	0.7500	0.0000
4 storeys	63.1	0.7500	0.0000
5 storeys	0.0000	2.03	203.0
8 storeys	0.0000	2.03	203.0
12 storeys	0.0000	2.03	203.0
20 storeys	0.0000	2.03	203.0

Table 3: Fire-resilience credit by compartmentation level

	Containment (%)	Loss freq (%)	Fire spread (bps)
Compartments: Not assessed	63.5	0.7400	4.50
Compartments: Partial	63.5	0.7400	5.00
Compartments: Meets minimum	66.7	0.6750	0.0000
Compartments: Enhanced	66.0	0.6900	0.0000
Compartments: Verified	66.3	0.6850	0.0000

Table 4: Fire-resilience credit by property condition

	Containment (%)	Loss freq (%)	Fire spread (bps)
Excellent	68.3	0.5250	0.0000
Good	67.0	0.6250	0.0000
Fair	66.7	0.6750	0.0000
Poor	66.5	0.7900	0.0000
Very poor	64.6	0.9700	0.0000

Table 5: Fire-resilience credit by commercial type

	Containment (%)	Loss freq (%)	Fire spread (bps)
Office	66.7	0.6750	0.0000
Retail	64.6	1.19	0.0000
Hotel	66.8	1.43	0.0000
Leisure	66.4	1.25	0.0000
Healthcare	65.4	0.9750	0.0000
MultiFamily	64.0	1.16	0.0000
MixedUse	66.4	1.52	0.0000
Other	66.4	1.52	0.0000

Table 6: Fire spread (bps) by height × suppression level — the fire-service-reach interaction

	Partial	Meets minimum	Enhanced	Verified
3 storeys	0.0000	0.0000	0.0000	0.0000
5 storeys	203.0	0.0000	0.0000	0.0000
8 storeys	203.0	0.0000	0.0000	0.0000
20 storeys	203.0	0.0000	0.0000	0.0000

Table 7: Fire-resilience credit by construction type / structural-frame combustibility (12-storey, sub-threshold 'Partial' suppression — above the fire-service reach, so material governs)

	Containment (%)	Loss freq (%)	Fire spread (bps)
Reinforced concrete	87.7	0.2500	0.0000
Concrete frame	83.5	0.3350	0.0000
Steel frame	78.8	0.4300	0.0000
Brick and block	78.8	0.4300	0.0000
Modern methods	63.8	0.7350	0.0000
Mixed construction	0.0000	2.03	203.0
Timber frame	0.0000	2.03	203.0

Table 8: Fire-resilience credit by automatic-detection level

	Containment (%)	Loss freq (%)	Fire spread (bps)
Detection: Not assessed	66.9	0.7150	0.0000
Detection: Partial	66.7	0.6750	0.0000
Detection: Meets minimum	66.7	0.6750	0.0000
Detection: Enhanced	66.7	0.6750	0.0000
Detection: Verified	67.3	0.6450	0.0000

Table 9: Fire-resilience credit by occupancy status (a Model A initiation-frequency driver)

	Containment (%)	Loss freq (%)	Fire spread (bps)
Fully occupied	66.7	0.6750	0.0000
Partially vacant	67.3	0.6150	0.0000
Vacant	67.4	0.5750	0.0000

Table 10: Fire spread (bps) by construction $\times$ height (sub-threshold 'Partial' suppression) — material, not height alone, gates the conflagration: non-combustible frames stay near zero at every height; combustible frames hit the reach cliff

	3 st	5 st	8 st	20 st
Reinforced concrete	0.0000	0.0000	0.0000	0.0000
Steel frame	0.0000	0.0000	0.0000	0.0000
Modern methods	0.0000	0.0000	0.0000	0.0000
Mixed construction	0.0000	203.0	203.0	203.0
Timber frame	3.50	203.0	203.0	203.0

Table 11: Containment (%) by construction $\times$ suppression level (12-storey, above the reach) — a non-combustible frame is contained by its structure even when suppression is weak; the two credits are complementary

	Partial	Meets minimum	Enhanced	Verified
Reinforced concrete	87.7	87.7	87.7	87.1
Steel frame	78.8	78.8	78.8	86.8
Modern methods	63.8	66.7	78.1	86.8
Mixed construction	0.0000	66.7	78.1	86.8
Timber frame	0.0000	66.5	78.1	86.8

Table 12: Construction sweep at *no* effective suppression (12-storey, 'Not assessed') — the robustness check. Even with suppression absent, a non-combustible frame realises effectively zero conflagration: the fuel cap holds because the chain absorbs before the capped intensity can creep across I_{crit} (FIRE-L3). The ceiling jitter leaves only a negligible tail (sub-bp at this baseline; single-digit bps in very tall non-combustible assets). Combustible frames suffer the full cliff.

	Containment (%)	Loss freq (%)	Fire spread (bps)
Reinforced concrete	87.5	0.2700	0.0000
Concrete frame	83.3	0.3600	0.0000
Steel frame	78.7	0.4600	0.0000
Brick and block	78.7	0.4600	0.0000
Modern methods	63.9	0.7800	0.0000
Mixed construction	0.0000	2.16	216.0
Timber frame	0.0000	2.16	216.0

Table 13: Config explainer: each **ConstructionType** maps to the model levers it sets — frame combustibility γ , the growth-rate scale, the fuel ceiling $I_{\max}$, the structural-containment blend floor w_{struct} , and the structural-fire-resistance fallback level. Deterministic (computed from `config/fire_matrices.json`).

Construction	γ	Growth scale	$I_{\max}$	w_{struct}	Struct. FR fallback
Reinforced concrete	0.0000	0.5000	5.00	1.00	Verified
Concrete frame	0.0500	0.5500	5.00	0.9500	Verified
Steel frame	0.1000	0.6000	5.00	0.9000	Enhanced
Brick and block	0.1000	0.6000	5.00	0.9000	Enhanced
Modern methods	0.2500	0.7500	5.00	0.7500	Meets minimum
Mixed construction	0.5000	1.00	37.9	0.5000	Partial
Timber frame	1.00	1.50	120.0	0.0000	Partial

2 Stress Testing

This section documents stress testing scenarios applied to the model under extreme but plausible conditions.

2.1 Stress Scenarios

Table 14: Stress testing scenarios for Building Fire-Resilience Credit Model.

Scenario	Parameter Shock	Severity	Status
Baseline	No shock	—	Passed
Moderate stress	+2 σ inputs	Medium	Pending
Severe stress	+3 σ inputs	High	Pending
Reverse stress	Output threshold breach	Critical	Pending

Detailed stress test results will be populated as scenarios are executed. Refer to the model’s validation plan for the full stress testing programme.

3 Model Performance

This section tracks key performance metrics for the model across validation and production environments.

Table 15: Performance metrics for Building Fire-Resilience Credit Model.

Metric	Value	Status
Unit test pass rate	See Test Results	—
Execution time (single run)	TBD	—
Numerical stability	TBD	—
Reproducibility	Deterministic	OK

This report is produced automatically; human review is required before formal model approval. Supporting artefacts are available in the model documentation directory.